

Incorrect Data Link Transmission from Remote Input/Output Unit to Diesel Control Unit

Symptoms

- No communication between the remote input/output unit and the DCU410 unit.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box wiring.	
	STEP 1A. Check the remote input/output unit Modicon™ communication bus supply and return wires for an open.	
	STEP 1B. Check the voltage at the remote input/output unit supply wire at the remote input/output unit.	

STEP 1. Check the customer interface box wiring.

STEP 1A. Check the remote input/output unit Modicon™ communication bus supply and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the remote input/output unit Modicon™ communication bus supply and return wires at the remote input/output unit and DCU410 unit.

Action	Specification/Repair	Next Step
Check the supply and return wires for an open • Place one test lead on the remote input/output Modicon™ communication bus supply wire at the DCU410 unit. • Place the other test lead on the remote input/output Modicon™ communication bus supply wire at the remote input/output unit. • Place one test lead on the remote input/output Modicon™ communication bus return wire at the DCU410 unit. • Place the other test lead on the remote input/output Modicon™ communication bus return wire at the remote input/output unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	1B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 1B. Check the voltage at the remote input/output unit supply wire at the remote input/output unit.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step
Check the voltage at the remote input/output Modicon™ communication bus supply wire. • Place one test lead on the remote input/output Modicon™ communication bus supply wire at the remote input/output unit. • Place the other test lead on the remote input/output Modicon™ communication bus return wire at the remote input/output unit. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC? YES Repair : Check or replace the batteries. Refer to the OEM service manual or contact a Cummins® Authorized Repair Location.	Repair complete
	Less than +24-VDC? NO	Repair complete